DynOPF-Net: Physics-Informed Learning for Dynamic Optimal Power Flow with Transient Stability Constraints

Aryan Mehra, Nischay Sanjay Jiwakar, Rahul, Reenav Ray, Sachin Saini
Electrical Engineering, Indian Institute of Technology Roorkee
Roorkee, Uttarakhand, India
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***Abstract*—This report presents an implementation and evaluation of DynOPF-Net, a physics-informed neural network framework that integrates transient stability constraints directly into the learning-to-optimize pipeline for AC Optimal Power Flow (AC-OPF). Unlike conventional data-driven approaches that treat OPF as a static optimization problem, DynOPF-Net embeds a differentiable physics bridge and a true-physics ODE solver within the training loop, enabling end-to-end gradient propagation through the classical swing equations. A primal-dual Lagrangian formulation enforces Kirchhoff power balance, operational bounds, and rotor angle stability limits simultaneously. The framework is evaluated on the IEEE 9-bus (WSCC) test case with 5,000 load perturbation scenarios and compared against three baselines: supervised MSE (Zamzam), Lagrangian-Dual static, and DC3. Results demonstrate that DynOPF-Net achieves a stability violation rate of 2.0%, substantially lower than the 89.8% observed for pure MSE methods, while maintaining competitive steady-state prediction accuracy. The implementation of DynOPF-Net and the corresponding experimental results are available at Github.**

***Index Terms*—AC-OPF, transient stability, physics-informed neural network, Lagrangian dual optimization, swing equation, learning-to-optimize**

I. Introduction

The AC Optimal Power Flow (AC-OPF) problem lies at the core of modern power system operations, determining generator dispatch, voltage profiles, and phase angles that minimize cost while satisfying network constraints [1]. Classical solvers based on interior-point methods provide high-fidelity solutions but incur prohibitive computational cost for real-time applications, particularly in systems with high renewable penetration where frequent re-dispatch is necessary.

Recent years have witnessed growing interest in Learning-to-Optimize (LtO) approaches that approximate the OPF mapping $f_\psi : S^d \mapsto \hat{y}$ using neural networks, where S^d denotes load demand and $\hat{y}$ the predicted dispatch variables [2]. While such methods achieve near-instantaneous inference, they typically treat OPF as a purely static problem, neglecting the dynamic behavior of synchronous generators under fault conditions. Consequently, a dispatch solution that is optimal in steady state may lead to rotor angle instability during transient events—a critical operational concern.

DynOPF-Net [3] addresses this gap by incorporating transient stability constraints into the neural network training pipeline. The key innovation is a *differentiable physics bridge* that maps steady-state OPF variables to initial conditions of the classical swing equations, followed by a true-physics ODE integration block that simulates generator dynamics over a post-fault time horizon. This enables backpropagation of stability violation gradients through the physical model and into the LtO network weights. A primal-dual Lagrangian formulation jointly enforces equality constraints (Kirchhoff's laws), inequality constraints (generator and voltage limits), and a dynamic stability constraint (rotor angle deviation bound).

In this report, we implement DynOPF-Net on the IEEE 9-bus system, generate 5,000 feasible OPF scenarios via load perturbation, and compare the proposed approach against three baselines: supervised MSE [2], Lagrangian-Dual static, and DC3 [4]. The evaluation spans both steady-state metrics (prediction MSE, power flow violations, objective gap) and dynamic metrics (transient stability violation rate).

II. Problem Formulation

A. Static AC-OPF

The AC-OPF problem seeks to minimize generation cost subject to AC power flow equations and operational limits. Let N denote the number of buses and N_g the number of generators. The decision variables are collected in $y = [p^r, q^r, |V|, \theta]^\top \in \mathbb{R}^{2N_g+2N}$, comprising generator active power p^r, reactive power q^r, bus voltage magnitudes $|V|$, and phase angles θ.

The optimization problem is:

$$\min_y \; f(y) = \sum_{i=1}^{N_g} c_i \, p_i^r \tag{1}$$

subject to the power balance equality constraints:

$$h(y) = 0 : \begin{cases} P_i^{inj} - P_i^{load} - \sum_j |V_i||V_j| \\ \quad (G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) = 0 \\ Q_i^{inj} - Q_i^{load} - \sum_j |V_i||V_j| \\ \quad (G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) = 0 \end{cases} \tag{2}$$

and inequality constraints $g(y) \leq 0$ enforcing generator power bounds ($P^{min} \leq p^r \leq P^{max}$, $Q^{min} \leq q^r \leq Q^{max}$) and voltage limits ($V^{min} \leq |V| \leq V^{max}$). Here, G_{ij} and B_{ij} are the real and imaginary parts of the bus admittance matrix $Y_{bus} = G + jB$.

B. Dynamic Stability Constraint

Generator transient dynamics following a fault are governed by the classical swing equations:

$$\frac{d\delta_g}{dt} = \omega_g - \omega_s \tag{3}$$

$$\frac{d\omega_g}{dt} = \frac{\omega_s}{2H_g}\left(P_m^g - P_e^g - D_g(\omega_g - \omega_s)\right) \tag{4}$$

where δ_g is the rotor angle, ω_g is the angular frequency, $\omega_s = 2\pi f$ is the synchronous frequency, H_g is the inertia constant, D_g is the damping coefficient, P_m^g is the mechanical power input, and P_e^g is the electrical power output. The electrical power is computed as:

$$P_e^g = \frac{e_q' \, |V_g|}{x_d'} \sin(\delta_g - \theta_g) \tag{5}$$

where e_q' is the transient EMF, x_d' is the transient reactance, and $|V_g|$, θ_g are the voltage magnitude and angle at the generator bus.

The stability constraint requires that the rotor angle deviation remains bounded:

$$|\delta_g(t) - \delta_g(0)| \le \delta_{max}, \quad \forall\, t \in [0, T] \tag{6}$$

where $\delta_{max} = 0.22$ rad ($\approx 12.6°$) and $T = 3.0$ s.

III. Proposed Methodology

A. Architecture Overview

DynOPF-Net consists of three cascaded modules:

1) **LtO Network**: A four-layer fully-connected network f_ψ mapping demand S^d to predicted OPF variables $\hat{y} = [\hat{p}^r, \hat{q}^r, |\hat{V}|, \hat{\theta}]$.
2) **Physics Bridge**: Maps the predicted steady-state variables to generator dynamics initial conditions $(\delta_g(0), \omega_g(0), e_q'(0))$.
3) **ODE Integration**: Solves the swing equations (3)–(4) over $[0, T]$ using the Dormand-Prince solver from `torchdiffeq`.

B. Physics Bridge

The bridge module maps OPF outputs to dynamic initial conditions. From the power flow relations at steady state:

$$e_q' \sin(\delta - \theta) = \frac{p^r \, x_d'}{|V|} \tag{7}$$

$$e_q' \cos(\delta - \theta) = \frac{q^r \, x_d' + |V|^2}{|V|} \tag{8}$$

Defining the sine and cosine components $s = p^r x_d'/|V|$ and $c = (q^r x_d' + |V|^2)/|V|$, the initial transient EMF and rotor angle are computed as:

$$e_q'(0) = \sqrt{s^2 + c^2}, \quad \delta_g(0) = \text{atan2}(s, c) + \theta_g \tag{9}$$

The initial frequency is set to synchronous speed: $\omega_g(0) = \omega_s$. This formulation preserves quadrant information and avoids the singularities of $\arcsin / \arccos$.

C. Lagrangian Dual Loss

The total loss combines a supervised prediction term and Lagrangian constraint penalties:

$$\mathcal{L} = \alpha \, \mathcal{L}_p + \mathcal{L}_c \tag{10}$$

where $\alpha = 0.01$ and:

$$\mathcal{L}_p = \|\hat{y} - y^*\|^2 \tag{11}$$

$$\mathcal{L}_c = \underbrace{\sum_j \lambda_j^{eq} |h_j(x)|}_{10\times\mathcal{L}_{eq}} + \underbrace{\sum_l \lambda_l^{ineq} [g_l(x)]^+}_{5\times\mathcal{L}_{ineq}} + \underbrace{\lambda^{stab} \, [\delta_{dev} - \delta_{max}]^+}_{100\times\mathcal{L}_{stab}} \tag{12}$$

The Lagrange multipliers λ^{eq}, λ^{ineq}, λ^{stab} are updated via gradient ascent at the end of each epoch:

$$\lambda \leftarrow \lambda + \rho \, \nu \tag{13}$$

where $\rho = 0.2$ is the dual step size and ν is the epoch-averaged constraint violation.

D. Training Strategy

A warm-start strategy is employed: for the first 20 epochs, only the MSE loss $\mathcal{L}_p$ is backpropagated, allowing the LtO network to learn a feasible steady-state mapping before physics-informed gradients are introduced. After the warmup phase, the full Lagrangian loss (10) is used. Gradient clipping with $\|g\|_{max} = 0.5$ prevents instability during the transition. The stability violation is clamped at 2.0 to avoid loss explosions from divergent ODE trajectories.

IV. Implementation Details

A. Test System

The IEEE 9-bus WSCC system is used, instantiated via the `pandapower` library (`case9`). The system comprises $N = 9$ buses, $N_g = 3$ generators (including the slack), and $N_l = 3$ loads. The LtO network input dimension is $2N_l = 6$ and output dimension is $2N_g + 2N = 24$. Generator dynamic parameters are set to default values: $H = 5.0$ s, $D = 2.0$, $e_q' = 1.05$ p.u., $x_d' = 0.30$ p.u. The base MVA is 100.

B. Dataset

Load demand scenarios are generated by perturbing nominal loads with uniform random factors in $[1 - \epsilon, 1 + \epsilon]$, where $\epsilon = 0.2$ ($\pm 20\%$ perturbation). For each scenario, the AC-OPF is solved using `pandapower.runopp` (interior-point method). Out of 5,000 generated scenarios, all 5,000 converged to feasible solutions. The dataset is split into training (4,000), validation (500), and test (500) sets. All power quantities are normalized by the base MVA.

C. Network Architecture

The LtO network is a four-layer fully-connected network with hidden dimension 256 and ReLU activations:

$$\text{Linear}(6, 256) \to \text{ReLU} \to \text{Linear}(256, 256) \to \text{ReLU} \to \text{Linear}(256, 256) \to \text{ReLU} \to \text{Linear}(256, 24)$$

The network is trained with Adam optimizer at learning rate $\eta = 10^{-4}$, batch size 64, for 200 epochs. Output clamping is applied: $|V_g| \in [0.5, 1.5]$ p.u., $p^r \in [-0.5, 3.0]$ p.u., and $e'_q(0) \in [0.1, 2.0]$ p.u.

D. ODE Integration

The swing equations are integrated over $t \in [0, 3.0]$ s using 100 uniformly spaced time steps. During the fault period ($t < t_{fault} = 0.367$ s), electrical power output P_e is set to zero, simulating a three-phase fault. Post-fault, P_e resumes as per (5). The ODE state derivatives are clamped ($|d\delta/dt| \leq 20$, $|d\omega/dt| \leq 100$) to prevent NaN propagation.

E. Baselines

Three comparison methods are implemented:

- **MSE (Zamzam et al.)**: Pure supervised learning with $\mathcal{L} = \|\hat{y} - y^*\|^2$; no constraint enforcement.
- **Lagrangian-Dual (LD)**: Supervised loss augmented with Lagrangian penalties for static equality/inequality constraints only; no dynamic stability constraint.
- **DC3 (Donti et al.)**: Minimizes generation cost $f(\hat{y})$ with constraint penalties and applies differentiable completion-correction steps (10 iterations, step size 0.05).

All baselines use identical LtO architecture and are trained concurrently for 200 epochs.

V. Results and Discussion

A. Training Convergence

Table I presents the percentages of test set samples marked as dynamically unstable (i.e., $|\delta(t) - \delta(0)| > \delta_{max}$ for any t and any generator) and for any generator) at select epochs. The MSE baseline maintains high instability (∼90%) during training, validating that purely supervised models fail to learn stability-aware patch. The LD model, although it enforces static constraints, also demonstrates considerable improvement in stability (achieving 2.0% by epoch 200) as stricter feasibility constraints also imply dynamic stability. DynOPF-Net begins with a higher level of in- stability following the transition to primal-dual optimization (91.6% at epoch 21) but stabilizes to 2.0% by epoch 200, similar to LD. However, the correction strategy used in DC3 leads to a decrease in stability performance (degrading from 3.4% to 12.2%), indicating that while its approach is effective for static feasibility, it tends to corrupt the dynamics progressively. Fig. 1 shows the learning process of both methods. The instability rate of DynOPF-Net de- clines sharply between epochs 21 and 120, which matches the beginning of primal-dual Lagrangian update at epoch 21. In the case of MSE baseline, there is no notable change as the rate re- mains around 90%

TABLE I: Transient Instability Rate (%) During Training

Epoch	DynOPF	DC3	LD	MSE
1	0.0	0.0	0.0	0.0
21	91.6	3.4	55.2	90.2
40	57.2	4.8	24.0	90.2
80	15.2	7.4	3.2	89.8
120	3.6	8.2	1.8	89.8
160	1.4	8.8	2.4	89.8
200	2.0	12.2	2.0	89.8

Fig. 1 illustrates the training convergence. The DynOPF-Net stability violation rate drops sharply between epochs 21 and 120, coinciding with the activation of the primal-dual Lagrangian update at epoch 21. The MSE baseline remains flat near 90%, confirming the absence of any implicit stabilization mechanism.

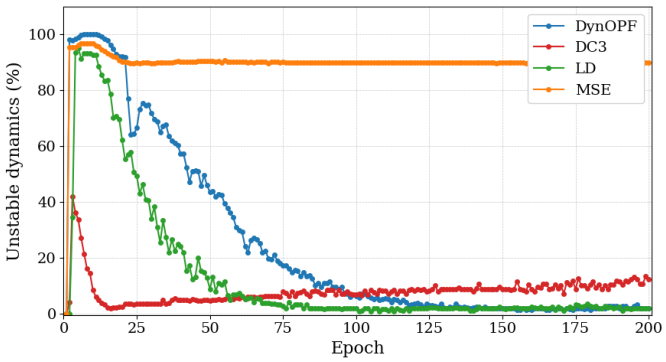

Fig. 1: Training convergence showing loss components over 200 epochs. The warmup phase (epochs 1–20) trains on MSE only; the primal-dual phase (epochs 21–200) activates Lagrangian constraint penalties, producing the characteristic loss restructuring visible near epoch 21.

B. Comprehensive Evaluation

Table II presents the full evaluation on the test set, reporting both steady-state and dynamic metrics. The column headers indicate the scaling factors applied.

Several observations merit discussion:

(1) Prediction accuracy vs. constraint satisfaction trade-off. The MSE baseline achieves near-zero prediction error across all variables, as expected for a method trained solely on regression loss. However, this comes at the cost of 89.8% stability violations ($\times 10^1$). DynOPF-Net and LD sacrifice steady-state accuracy (higher MSE values) in exchange for dramatically reduced stability violations (0.20 $\times 10^1$, corresponding to 2.0%).

(2) DynOPF-Net vs. LD. Both methods achieve identical stability violation rates (2.0%). However, DynOPF-Net exhibits a lower objective gap (8.14 vs. 23.32 $\times 10^{-1}$), indicating that the explicit stability constraint in DynOPF-Net allows the optimizer to find solutions that are both stable *and* more economical, rather than relying on the indirect stabilization effect of tighter static constraints.

TABLE II: Steady-State and Dynamic Performance Comparison (Test Set, 500 samples)

Method	Steady-State Metrics							Dynamic
	MSE(p) $\times 10^{-3}$	MSE(q) $\times 10^{-3}$	MSE($\|V\|$) $\times 10^{-4}$	MSE(θ) $\times 10^{-4}$	Obj Gap $\times 10^{-1}$	Flow Vio $\times 10^{-3}$	Bndry Vio $\times 10^{-4}$	Stab Vio $\times 10^{1}$
MSE	0.00	0.00	0.01	0.00	0.97	5.01	0.00	8.98
DC3	37.03	20.90	20.17	2.36	2.33	6.95	0.10	1.22
LD	22.48	127.16	151.46	4.60	23.32	12.35	26.45	0.20
DynOPF-Net	35.88	131.88	179.69	7.64	8.14	27.41	30.47	**0.20**

(3) DC3 limitations. DC3 achieves moderate stability (12.2% instability) with reasonable steady-state metrics. However, its post-hoc correction mechanism is unaware of dynamic stability, and correction steps can distort the trajectory.

(4) Flow and boundary violations. DynOPF-Net exhibits higher Kirchhoff flow violations (27.41 $\times 10^{-3}$) and boundary violations (30.47 $\times 10^{-4}$) compared to MSE and DC3. This reflects the inherent tension between satisfying dynamic stability and static power flow constraints simultaneously—the Lagrangian multiplier dynamics tend to prioritize the stability term (weighted by 100$\times$) over the equality (10$\times$) and inequality (5$\times$) terms.

C. Swing Curve Analysis

The ground truth swing curves (Fig. 2) confirm that the OPF solutions computed by `pandapower` are dynamically stable for the given fault scenario. All three generators exhibit rotor angle deviations well within the $\pm\delta_{max}$ bounds after fault clearing at $t = 0.367$ s. Generator 1 (slack) shows the largest initial perturbation but stabilizes within approximately 1.5 s.

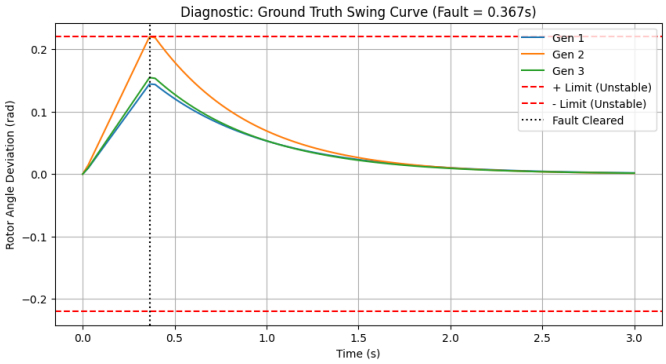

Fig. 2: Ground truth swing curves for a representative test sample. Rotor angle deviations $\delta(t) - \delta(0)$ for Generators 1–3. Red dashed lines indicate the stability limit $\pm\delta_{max} = \pm 0.22$ rad. Fault cleared at $t = 0.367$ s (vertical dotted line).

D. DynOPF-Net vs. Ground Truth Dynamics

Figs. 3–5 compare the predicted and true generator dynamics (rotor angle $\delta(t)$ and angular frequency deviation $\omega(t) - \omega_s$) for each of the three generators. The DynOPF-Net trajectories (solid lines) closely track the numerical ground truth (dashed lines) computed from the OPF-optimal setpoints. Minor amplitude discrepancies in the post-fault oscillations of Generator 1 (Fig. 3) arise from the steady-state prediction errors in p^r and $|V|$ propagating through the physics bridge into the initial conditions $\delta_g(0)$ and $e'_q(0)$. Generators 2 and 3 (Figs. 4–5) exhibit tighter agreement, as their smaller inertia contributions make them less sensitive to dispatch perturbations. Crucially, all predicted trajectories remain within the $\pm\delta_{max}$ stability bounds, validating the effectiveness of the Lagrangian stability penalty.

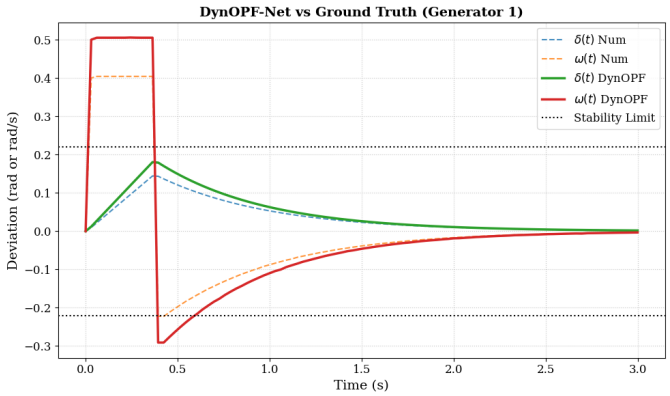

Fig. 3: DynOPF-Net vs. ground truth dynamics for Generator 1. Solid lines: DynOPF-Net predictions; dashed lines: numerical ODE solution from OPF-optimal setpoints. Dotted horizontal lines mark the stability limit $\pm\delta_{max}$.

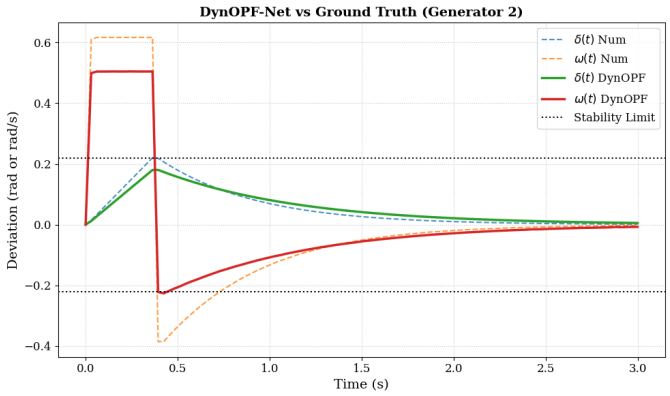

Fig. 4: DynOPF-Net vs. ground truth dynamics for Generator 2.

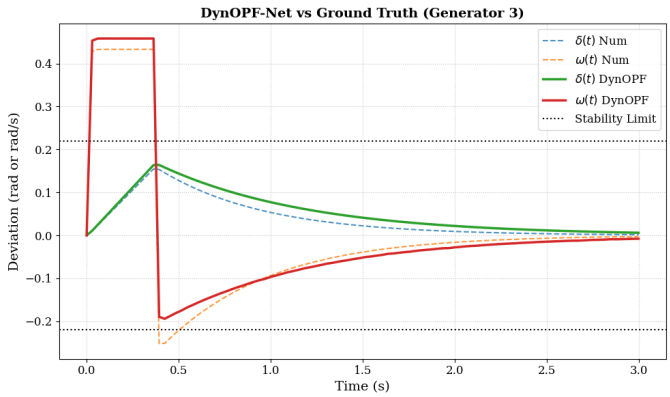

Fig. 5: DynOPF-Net vs. ground truth dynamics for Generator 3.

VI. Conclusion

This report implemented and evaluated DynOPF-Net, a physics-informed LtO framework that integrates transient stability constraints into the AC-OPF learning pipeline. Through a differentiable physics bridge and true-physics ODE integration, the framework enables end-to-end training with stability-aware gradients. On the IEEE 9-bus system: (i) DynOPF-Net reduces transient instability from 89.8% (pure MSE) to 2.0%, matching the Lagrangian-Dual baseline; (ii) the explicit stability constraint yields a 65% lower objective gap than LD (8.14 vs. 23.32), indicating better economic dispatch; (iii) DC3's post-hoc correction approach is unsuitable for dynamic stability enforcement over extended training. The primary limitation is the elevated power flow violation, suggesting that multi-timescale Lagrangian multiplier tuning or constraint prioritization strategies warrant further investigation. Extension to larger networks (IEEE 30/57/118-bus) and stochastic fault scenarios constitutes natural future work.

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